

SPL 2025 - Day 4

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Exercise 1 (Basic Statistics)

a) Read the General Social Survey (slide 22)

```
library(rio)
durl <- "https://gss.norc.oregonstate.edu/Documents/stata/"
gss <- import(paste0(durl, "2018_stata.zip"))
```

b) Create appropriate plots

We attach the data set to access the variables directly, which is not problematic in case (b).

However, note that in general one has to be careful when using **attach**: changes in the data set are **not applied automatically**.

That is, if we modify a variable, we have to attach the data set again; otherwise, the previous (non-modified) version of the variable will still be used whenever it is accessed directly.

To avoid a very long message about masked objects (which would list all variables of **gss**, since it is already attached), we will first **detach()** the data set whenever we change variables. This will be the case later, e.g., c).

```
attach(gss)
```

```
## The following object is masked from package:base:
```

```
##
```

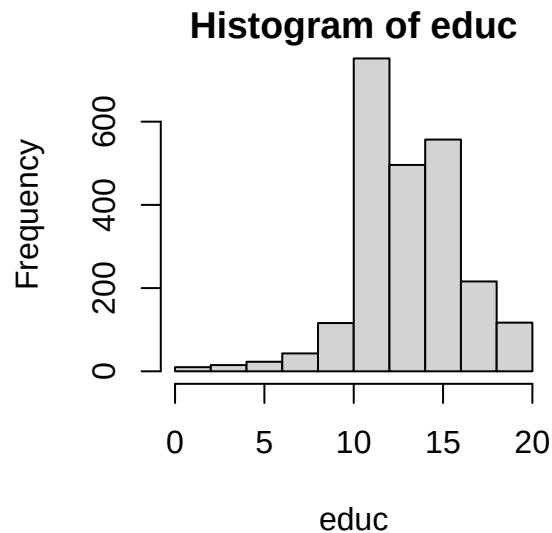
```
##      version
```

```
par(mfrow = c(1, 2), mar = c(5, 4, 1, 2) + 0.1)
```

```
# "mar" controls the margins of the plot (order: below, left, above, right) with default
# mar = c(5, 4, 4, 2) + 0.1; We don't have titles, i.e., we don't need much space above
# the plots (higher respective mar-value would lead to empty space)
```

```
hist(sibs)
```

```
hist(educ)
```



```
par(mfrow = c(1, 1))
summary(sibs)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   NA's
##      0.00   2.00   3.00   3.15   5.00   6.00     5
```

```
summary(educ)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   NA's
##      0.00  12.00  14.00  13.73  16.00  20.00     3
```

- Both distributions are quite skewed.
- A good measure of location is the **median**.
- To obtain a good summary, one should consider multiple statistics.

c) Consider income (rincome)

```
str(rincome)
```

(i) Examine the variable using the function str()

```
## num [1:2348] NA NA 12 12 NA NA 10 12 2 12 ...
## - attr(*, "label")= chr "respondents income"
## - attr(*, "format.stata")= chr "%29.0g"
## - attr(*, "labels")= Named num [1:24] 1 2 3 4 5 6 7 8 9 10 ...
## ..- attr(*, "names")= chr [1:24] "under $1,000" "$1,000 to $2,999" "$3,000 to $3,999" "$4,000 to $"
```

The variable has class **numeric** rather than **factor**. However, it also has several attributes.

The last attribute, "labels", itself has an additional attribute "names", i.e., it is a named vector.

These names provide reasonable labels when transforming the variable to class **factor**.

Note that only the first 12 labels appear in the vector (see `table(rincome)` below); therefore, we only use the first 12 names as labels.

Dig deeper: execute `names(rincome)`. Do you obtain vector of names? Most likely not – sometimes `str()` shows that a variable has a "names" attribute, but calling `names()` directly returns **NULL**.

This happens when the "names" are not attached to the variable itself, but to one of its attributes (often "labels"). Compare:

```
names(rincome)
```

```
## NULL
```

```
names(attr(rincome, "labels"))
```

```
## [1] "under $1,000"           "$1,000 to $2,999"
## [3] "$3,000 to $3,999"       "$4,000 to $4,999"
## [5] "$5,000 to $5,999"       "$6,000 to $6,999"
## [7] "$7,000 to $7,999"       "$8,000 to $9,999"
## [9] "$10,000 to $14,999"     "$15,000 to $19,999"
## [11] "$20,000 to $24,999"     "$25,000 or more"
## [13] "don't know"             "iap"
## [15] "I don't have a job"      "dk, na, iap"
## [17] "no answer"              "not imputable"
## [19] "refused"                "skipped on web"
## [21] "uncodeable"             "not available in this release"
## [23] "not available in this year" "see codebook"
```

```
attr(rincome, "labels")
```

```
##           under $1,000           $1,000 to $2,999
##                1                2
##           $3,000 to $3,999       $4,000 to $4,999
##                3                4
##           $5,000 to $5,999       $6,000 to $6,999
##                5                6
##           $7,000 to $7,999       $8,000 to $9,999
##                7                8
##           $10,000 to $14,999     $15,000 to $19,999
##                9               10
##           $20,000 to $24,999     $25,000 or more
##               11               12
##           don't know             iap
##                NA              NA
##           I don't have a job      dk, na, iap
##                NA              NA
##                no answer         not imputable
##                NA              NA
##                refused           skipped on web
##                NA              NA
##           uncodeable not available in this release
##                NA              NA
##           not available in this year      see codebook
##                NA              NA
```

```
table(rincome)
```

```
## rincome
##  1  2  3  4  5  6  7  8  9 10 11 12
## 33 32 32 21 21 12 18 33 94 61 107 899
```

```
detach(gss) # detach data before making changes
gss$rincome <- factor(gss$rincome,
                     labels = names(attr(gss$rincome, "labels"))[1:12])
```

```
# Alternative:
```

```
# attr(gss$rincome, "labels") <- attr(gss$rincome, "labels")[1:12]
```

```
# gss$rincome <- factorize(gss$rincome)
```

```
# rincmdol -- respondent's income in US$ where you replace the  
# category value by the class center, e.g. $1,000 to $2,999 by the numeric value 2000.  
# For the last category use 25000.
```

```
# New vector  
gss$rincmdol <- gss$rincome  
levels(gss$rincmdol)
```

(ii) Create a numeric variable rincmdol

```
## [1] "under $1,000"      "$1,000 to $2,999"   "$3,000 to $3,999"  
## [4] "$4,000 to $4,999"   "$5,000 to $5,999"   "$6,000 to $6,999"  
## [7] "$7,000 to $7,999"   "$8,000 to $9,999"   "$10,000 to $14,999"  
## [10] "$15,000 to $19,999" "$20,000 to $24,999" "$25,000 or more"
```

```
#####  
# Add centres manually ###  
#####  
levels(gss$rincmdol) <- c("500", "2000", "3500", "4500", "5500", "6500",  
                          "7500", "9000", "12500", "17500", "22500", "25000")
```

```
# Try first as.numeric directly: not successful coercion
```

```
gss$rincmdol <- as.numeric(as.character(gss$rincmdol))  
# as.character accesses the labels  
# as.numeric transforms it to numeric  
attach(gss) # attach updated data
```

```
## The following object is masked from package:base:
```

```
##
```

```
## version
```

```
#####  
# Add centers by coding #  
#####
```

```
# Remove the thousands separator ","  
# gsub replaces "," with "" several times  
# levels(gss$rincmdol) <- gsub(",", "", levels(gss$rincmdol))  
# levels(gss$rincmdol)
```

```
# Extract numbers from labels and convert them to numeric  
# regmatches extracts matched substrings  
# greexpr returns the position of ea in the lines  
# boundaries <- regmatches(levels(gss$rincmdol), greexpr("[0-9]+",  
# levels(gss$rincmdol)))  
# boundaries
```

```
# Change to numeric  
# boundaries <- lapply(boundaries, as.numeric)  
# boundaries  
# Compute centers
```

```
# centers <- c(boundaries[[1]]/2, # left boundary corresponds to 0
#             sapply(boundaries[2:11], function(x) round(mean(x))),
#             boundaries[[12]]) # last value is just the value itself
# centers
# levels(gss$rincmdol) <- centers

# attach(gss) # attach updated data
```

```
# ?aggregate
# Compute Summary Statistics of Data Subsets
aggregate(rincmdol, by = list(degree), FUN = mean, na.rm = TRUE)
```

(iii) Compute the average respondent's income

```
##      Group.1      x
## 1      0 16117.12
## 2      1 19453.83
## 3      2 20685.61
## 4      3 22821.09
## 5      4 23476.19
```

```
# Alternative: tapply(rincmdol, degree, mean, na.rm = TRUE)
```

```
mean(rincmdol, na.rm = TRUE)
```

```
## [1] 20570.43
```

```
attr(gss$degree, "labels")
```

```
##      less than high school      high school
##      0                      1
##      associate/junior college      bachelor's
##      2                      3
##      graduate                  don't know
##      4                      NA
##      iap                      I don't have a job
##      NA                      NA
##      dk, na, iap              no answer
##      NA                      NA
##      not imputable            refused
##      NA                      NA
##      skipped on web          uncodeable
##      NA                      NA
## not available in this release  not available in this year
##      NA                      NA
##      see codebook
##      NA
```

The average income for all respondents is 20570.43, for people with less than a high school degree 16117.12 and for people with a graduate degree 23476.19.

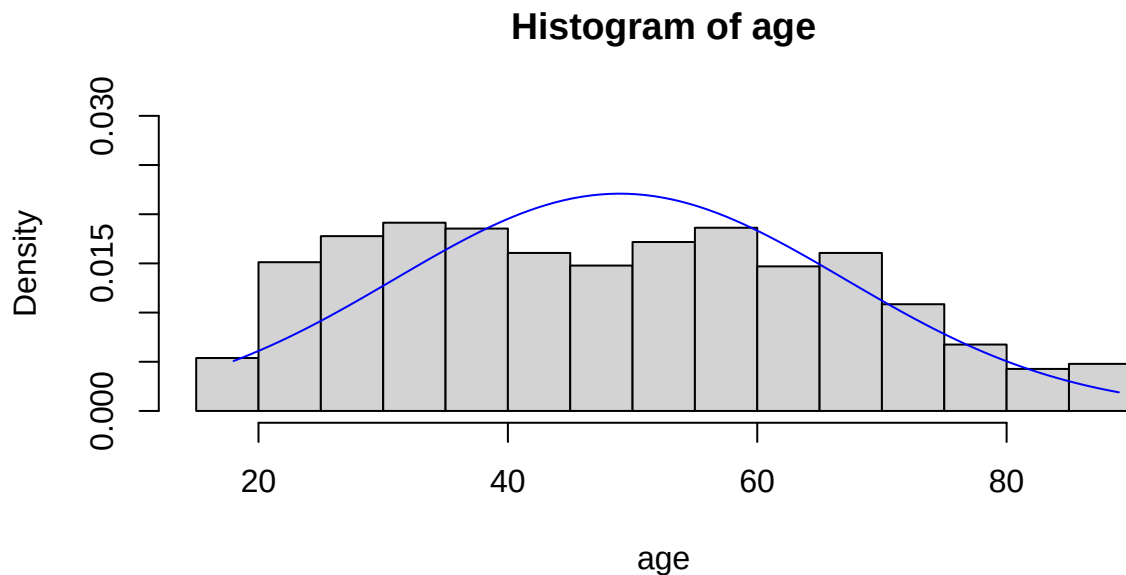
(iv) Do you think that staying in school pays off? The mean increases with degree of education, so yes.

d) Consider the variable `age`.

(i) create a histogram with total area

```
hist(age, freq = FALSE, ylim = c(0, 0.03))
m <- mean(age, na.rm = TRUE)
sd_dat <- sd(age, na.rm = TRUE)
seq <- seq(min(age, na.rm = TRUE), max(age, na.rm = TRUE), length.out = 100)
lines(seq, dnorm(x = seq, mean = m, sd = sd_dat), col = "blue")
```

(ii) overlay the histogram with the density of an appropriate normal distribution. Does it look like normally distributed?



Does not seem normally distributed, looks rather bimodal.

Key point for students: `lines()`/`points()`/`abline()` must run **after the initial plot in the same device** (i.e., the same chunk when knitting).

e) Analyse the variable `age` with boxplots grouped by `life`.

The transformation to `factor` is analogous to that of `rincome` in exercise c) (i).

Here, only the first 3 labels appear in the vector; therefore, we only use the first 3 names as labels.

```
str(life)

##  num [1:2348] 2 1 NA NA 2 2 2 3 NA NA ...
## - attr(*, "label")= chr "is life exciting or dull"
## - attr(*, "format.stata")= chr "%29.0g"
## - attr(*, "labels")= Named num [1:15] 1 2 3 NA NA NA NA NA NA NA ...
## ..- attr(*, "names")= chr [1:15] "exciting" "routine" "dull" "don't know" ...
```

```
table(life)
```

```
## life
##   1   2   3
## 766 717  72
```

```
detach(gss) # detach data before making changes
gss$life <- factor(gss$life, labels = names(attr(gss$life, "labels"))[1:3])
```

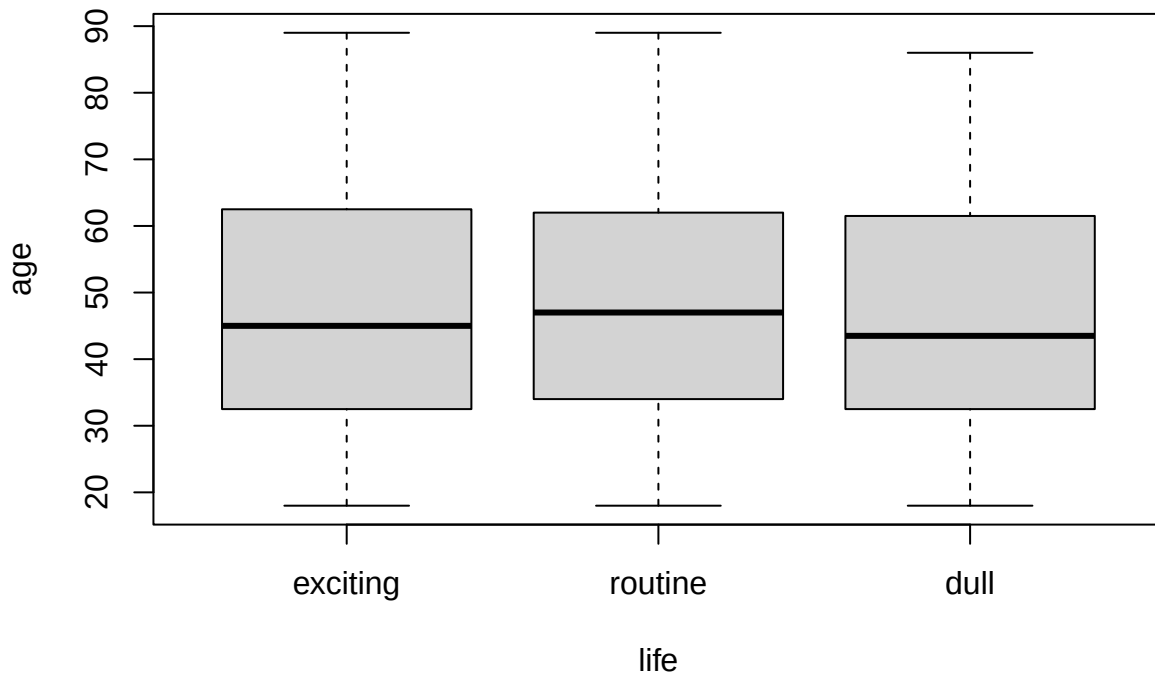
```
# Alternative:
# attr(gss$life, "labels") <- attr(gss$life, "labels")[1:3]
# gss$life <- factorize(gss$life)
attach(gss) # attach updated data
```

```
## The following object is masked from package:base:
```

```
##
```

```
## version
```

```
boxplot(age ~ life, data = gss)
```



```
aggregate(age, by = list(life), sd, na.rm = TRUE)
```

(i) Which group has the largest dispersion? Which criterion do you use for dispersion?

```
## Group.1 x
## 1 exciting 18.30171
## 2 routine 18.25461
## 3 dull 18.09502
```

```
# Alternative: tapply(age, life, sd, na.rm = TRUE)
```

Using the standard deviation as criterion for dispersion, the first group (“exciting”) has the largest dispersion. Other criteria (e.g., variance, range, interquartile range, coefficient of variation) can be used, as well.

(ii) Is the distribution in one or more groups symmetric? Based on the boxplots, none of the distributions looks symmetric.

```
aggregate(age, by = list(life), median, na.rm = TRUE)
```

(iii)

```
##      Group.1      x
## 1 exciting 45.0
## 2 routine 47.0
## 3      dull 43.5

#Alternative: tapply(age, life, median, na.rm = TRUE)
```

(iv) It is difficult to infer anything from this plot – further analysis is needed.

f) Consider the relationship between how happy a person is in general (happy) and the happiness of his or her marriage (hapmar).

```
str(happy)
```

(i) Transform both variables to class factor and create a Mosaic plot

```
## num [1:2348] 2 1 1 1 2 3 2 2 3 2 ...
## - attr(*, "label")= chr "general happiness"
## - attr(*, "format.stata")= chr "%29.0g"
## - attr(*, "labels")= Named num [1:15] 1 2 3 NA NA NA NA NA NA NA ...
## ..- attr(*, "names")= chr [1:15] "very happy" "pretty happy" "not too happy" "don't know" ...
```

```
table(happy)
```

```
## happy
##      1      2      3
## 701 1307  336
```

```
detach(gss) # detach data before making changes
gss$happy <- factor(gss$happy, labels = names(attr(gss$happy, "labels"))[1:3])
```

```
# Alternative:
# attr(gss$happy, "labels") <- attr(gss$happy, "labels")[1:3]
# gss$happy <- factorize(gss$happy)
```

```
str(gss$hapmar)
```

```
## num [1:2348] NA NA 1 1 NA NA NA NA NA 2 ...
## - attr(*, "label")= chr "happiness of marriage"
## - attr(*, "format.stata")= chr "%29.0g"
## - attr(*, "labels")= Named num [1:15] 1 2 3 NA NA NA NA NA NA NA ...
## ..- attr(*, "names")= chr [1:15] "very happy" "pretty happy" "not too happy" "don't know" ...
```

```
table(gss$hapmar)
```

```
##
##      1      2      3
## 638 324  30
```

```
gss$hapmar <- factor(gss$hapmar, labels = names(attr(gss$hapmar, "labels"))[1:3])
```

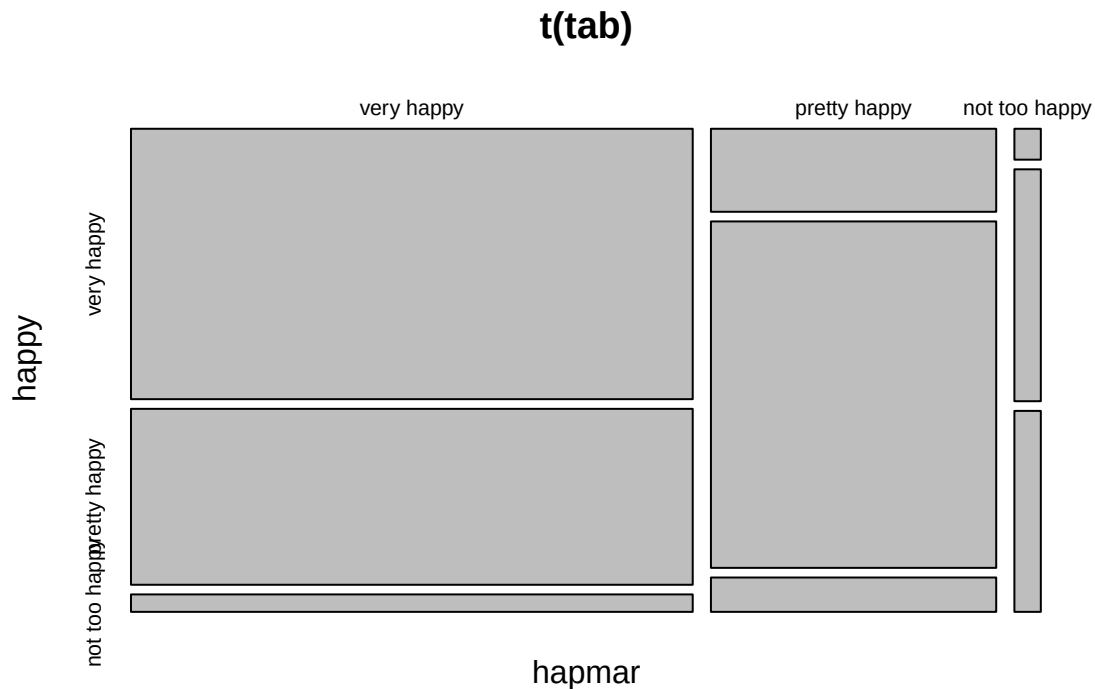
```
# Alternative:
# attr(gss$hapmar, "labels") <- attr(gss$hapmar, "labels")[1:3]
# gss$hapmar <- factorize(gss$hapmar)
```

```
attach(gss) # attach updated data
```



```
## The following object is masked from package:base:
##
##      version
# Mosaic plot recap:
# width of bars: proportional to margin frequencies  $P(X = x)$ 
# height of bars: proportional to cond. frequencies  $P(Y = y|X = x)$ 
# area: prop to joint

tab <- table(happy, hapmar)
plot(t(tab))
```



happy and hapmar are not independent.

```
ctab <- table(happy, hapmar)
ctab
```

(ii) Perform a chi square independence test.

```
##           hapmar
## happy      very happy pretty happy not too happy
## very happy      372      58         2
## pretty happy     242     242        15
## not too happy     24      24        13
```

```
chisq_res <- chisq.test(ctab)
```

```
## Warning in chisq.test(ctab): Chi-squared approximation may be incorrect
```

```
chisq_res
```

```
##
## Pearson's Chi-squared test
##
## data:  ctab
```

```
## X-squared = 222.15, df = 4, p-value < 2.2e-16
# expected counts in chi-squared  $E_{ij} = (\text{row total}_i)(\text{col total}_j)/N$ 
# test stat  $V = \text{sum}((\text{observed} - \text{expected})^2 / \text{expected})$ 
chisq_res$exp
```

```
##                hapmar
## happy          very happy pretty happy not too happy
## very happy    277.83871   141.09677   13.064516
## pretty happy  320.92944   162.97984   15.090726
## not too happy  39.23185    19.92339    1.844758
```

```
# run Fisher's exact test
fisher_res <- fisher.test(ctab)
fisher_res
```

```
##
## Fisher's Exact Test for Count Data
##
## data: ctab
## p-value < 2.2e-16
## alternative hypothesis: two.sided
```

Chi-squared test: $p\text{-value} < 0.0001 < 0.05 \rightarrow$ the null hypothesis of independence of **happy** and **hapmar** can be rejected, i.e., they do not appear to be independent. However, as the rule of thumb that all expected counts are at least 5 is not fulfilled, the test might not be appropriate for this table. An alternative test, Fisher's exact test, also rejects the null hypothesis of independence with $p\text{-value} < 0.0001 < 0.05$.

```
library(DescTools)

# Cramer's Phi: range -1 to 1
Phi(happy, hapmar)
```

(iii) Compute Cramers Φ , V and the contingency coefficients.

```
## [1] 0.4732244
```

```
# Cramer's V : range 0 to 1
CramerV(happy, hapmar)
```

```
## [1] 0.3346202
```

```
# contingency coeff : 0 to 1
ContCoef(happy, hapmar)
```

```
## [1] 0.4277469
```

```
# corrected contingency coeff: 0 1
ContCoef(happy, hapmar, correct = TRUE)
```

```
## [1] 0.5238808
```

All values suggest a moderate association between **happy** and **hapmar**.

g) Compute the correlation between **paeduc** and **maeduc**.

```
# Kendall's : rank association between concordant and discordant pairs
cor(paeduc, maeduc, use = "complete.obs", method = "kendall")
```

```
## [1] 0.534485
```

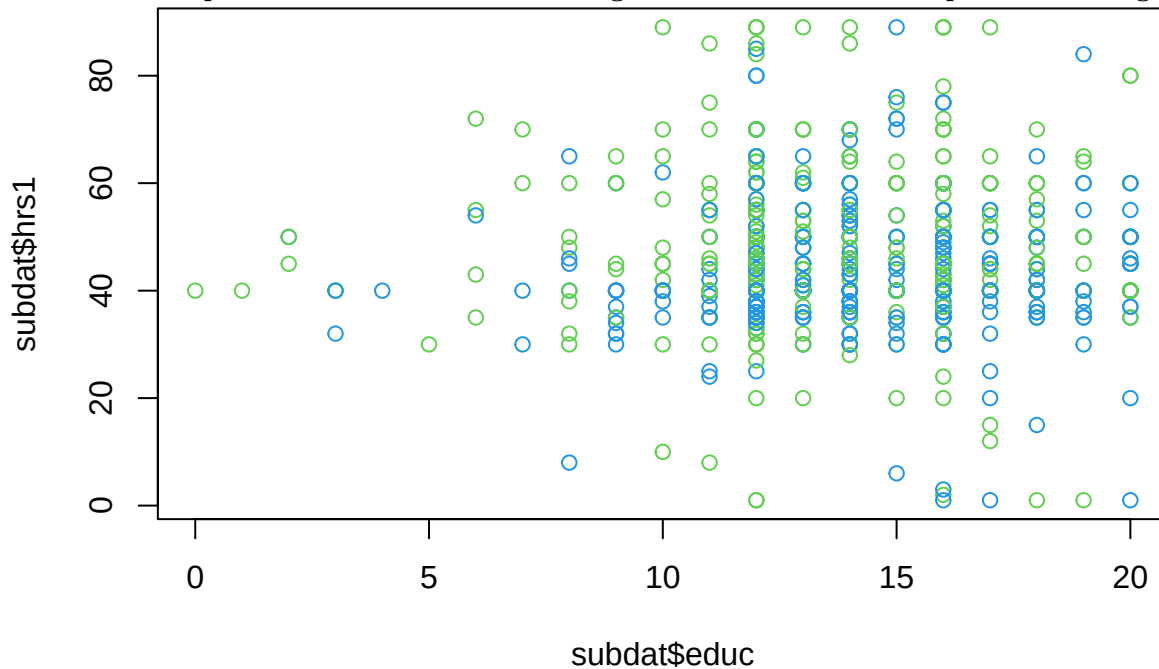
The number of years of education seems to be moderately correlated.

h) Select full-time workers only (`wrkstat == 1`)

```
subdat <- gss[wrkstat == 1, ]
```

```
plot(subdat$educ, subdat$hrs1, col = (subdat$sex + 2))
```

(i) Create a scatter plot with `hrs1` and `educ`. Change the colour of the data points according to



`sex`

The argument `col = subdat$sex` tells R to use the values of `sex` to assign colours.

The variable `subdat$sex` takes only two values (say 1 for male, 2 for female).

R interprets these numbers as colour codes:

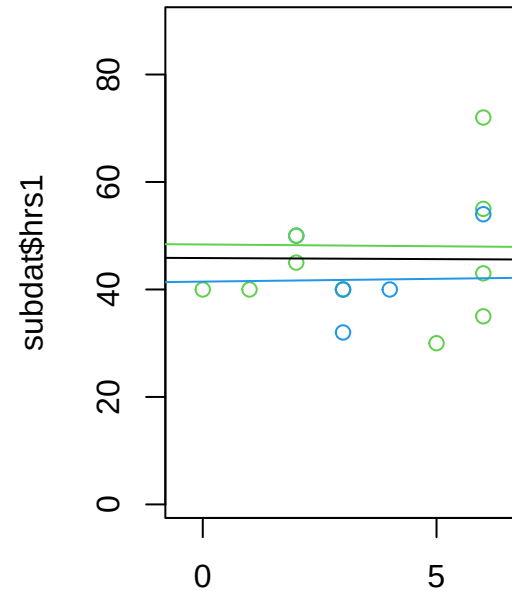
- 1 -> the first colour in R's default palette (usually black)
- 2 -> the second colour (usually red)

I added +2 to 1 and 2 to have more vibrant colours.

```
lms <- list(  
  "both" = lm(hrs1 ~ educ, data = subdat),  
  "male" = lm(hrs1 ~ educ, data = subset(subdat, sex == 1)),  
  "female" = lm(hrs1 ~ educ, data = subset(subdat, sex == 2))  
)  
  
# Alternative:  
# lms <- lapply(list("both" = subdat,  
#                   "male" = subset(subdat, sex == 1),
```

```
#           "female" = subset(subdat, sex == 2)),
#           lm, formula = hrs1 ~ educ)

# (iii)
plot(subdat$educ, subdat$hrs1, col = (subdat$sex + 2))
abline(lms$both, col = "black")
abline(lms$male, col = 3)
abline(lms$female, col = 4)
```



(ii) Run simple linear regressions for both sexes together and separately

i) Test whether Clinton and Trump supporters (pres16) differ in age and education

```
table(pres16)
```

```
## pres16
##   1   2   3   4
## 764 577  87  20
```

```
attr(pres16, "labels")
```

```
##               clinton               trump
##                1                2
##   other candidate   didn't vote for president
##                3                4
##   don't know              iap
##                NA              NA
##   I don't have a job      dk, na, iap
##                NA              NA
##   no answer              not imputable
##                NA              NA
##   refused                skipped on web
##                NA              NA
##   uncodeable not available in this release
##                NA              NA
```

```
##      not available in this year          see codebook
##                                NA                                NA
# Selected those who voted for Trump or Clinton: pres16 = 1 or 2
ind <- pres16 %in% c(1, 2)
t.test(age[ind] ~ pres16[ind])

##
## Welch Two Sample t-test
##
## data:  age[ind] by pres16[ind]
## t = -4.6405, df = 1297.7, p-value = 3.827e-06
## alternative hypothesis: true difference in means between group 1 and group 2 is not equal to 0
## 95 percent confidence interval:
##  -6.132427 -2.488089
## sample estimates:
## mean in group 1 mean in group 2
##      52.17585      56.48611

# Runs a two-sample t-test comparing mean age between Clinton (pres16 = 1)
# and Trump (pres16 = 2) supporters.
t.test(educ[ind] ~ pres16[ind])

##
## Welch Two Sample t-test
##
## data:  educ[ind] by pres16[ind]
## t = 2.8418, df = 1299.6, p-value = 0.004556
## alternative hypothesis: true difference in means between group 1 and group 2 is not equal to 0
## 95 percent confidence interval:
##  0.1313707 0.7171099
## sample estimates:
## mean in group 1 mean in group 2
##      14.50916      14.08492

# The same for age
```

Both p -values are smaller than 0.05, i.e., in both tests the null hypothesis of identical means can be rejected. Thus, we conclude that Clinton and Trump supporters differ in average age and education.

j) Use t-test to identify possible differences of averages in age, educ tvhours for different groups in ‘richwork’

```
table(richwork)

## richwork
##    1    2
## 723 280

t.test(age ~ richwork)

##
## Welch Two Sample t-test
##
## data:  age by richwork
## t = -6.4568, df = 519.31, p-value = 2.46e-10
## alternative hypothesis: true difference in means between group 1 and group 2 is not equal to 0
```

```
## 95 percent confidence interval:
## -8.040245 -4.288950
## sample estimates:
## mean in group 1 mean in group 2
## 40.79612 46.96071
# 2.46e-10 < 0.05 => null hypothesis of identical means can be rejected
t.test(educ ~ richwork)

##
## Welch Two Sample t-test
##
## data: educ by richwork
## t = -1.0719, df = 527.79, p-value = 0.2843
## alternative hypothesis: true difference in means between group 1 and group 2 is not equal to 0
## 95 percent confidence interval:
## -0.6050930 0.1778846
## sample estimates:
## mean in group 1 mean in group 2
## 13.89751 14.11111
# 0.2843 > 0.05 => null hypothesis of identical means can not be rejected
t.test(tvhours ~ richwork)

##
## Welch Two Sample t-test
##
## data: tvhours by richwork
## t = -2.3347, df = 332.85, p-value = 0.02016
## alternative hypothesis: true difference in means between group 1 and group 2 is not equal to 0
## 95 percent confidence interval:
## -0.74533651 -0.06368065
## sample estimates:
## mean in group 1 mean in group 2
## 2.137427 2.541935
# 0.02016 < 0.05 => null hypothesis of identical means can be rejected
```

k) Run a regression model

```
m <- lm(educ ~ speduc + paeduc + rincmdol + tvhours, data = gss)
```

In order to check the normality assumption, one can, e.g., use a Normal Q-Q plot:

```
plot(m, which = 2)
```

